

DESIGN AND IMPLEMENTATION OF 400 V DISTRIBUTION LINE WITH SMART STREET LIGHT CONTROL SYSTEM

This project designs a 400 V low-voltage distribution system with automated street-light sensor control to improve energy efficiency and grid reliability.

Aim

- Design and construct a 400 V distribution line system equipped with smart street light sensors.

Objectives

- Design low-voltage (400V) overhead distribution line according to standard electrical engineering.
- Implement light sensors (LDR/Photocell) for automatic dusk-to-dawn street lighting control.
- Reduce energy consumption, operational costs, and manual intervention in street light management.

Methodology

Load data collection → Distribution line calculations → Equipment selection → Sensor integration and testing → Performance evaluation.



HYBRID SOLAR POWER SYSTEM FOR DOMESTIC LOAD AND STREET LIGHTING

This project presents a hybrid solar power system using solar energy and battery storage to reliably supply domestic load and street lighting.

Aim

- To design and implement a hybrid solar power system for domestic load and street lighting.

Objectives

- To design the solar system according to the required load and select suitable solar panels, batteries, inverter, and charge controller.
- To provide a reliable and continuous power supply for the required loads.
- To reduce electricity consumption and operating costs while improving energy efficiency through renewable solar energy.

Methodology

Load Calculation → Solar PV Sizing → Component Selection → System Design → Circuit Connection → Simulation/Testing → Result Evaluation



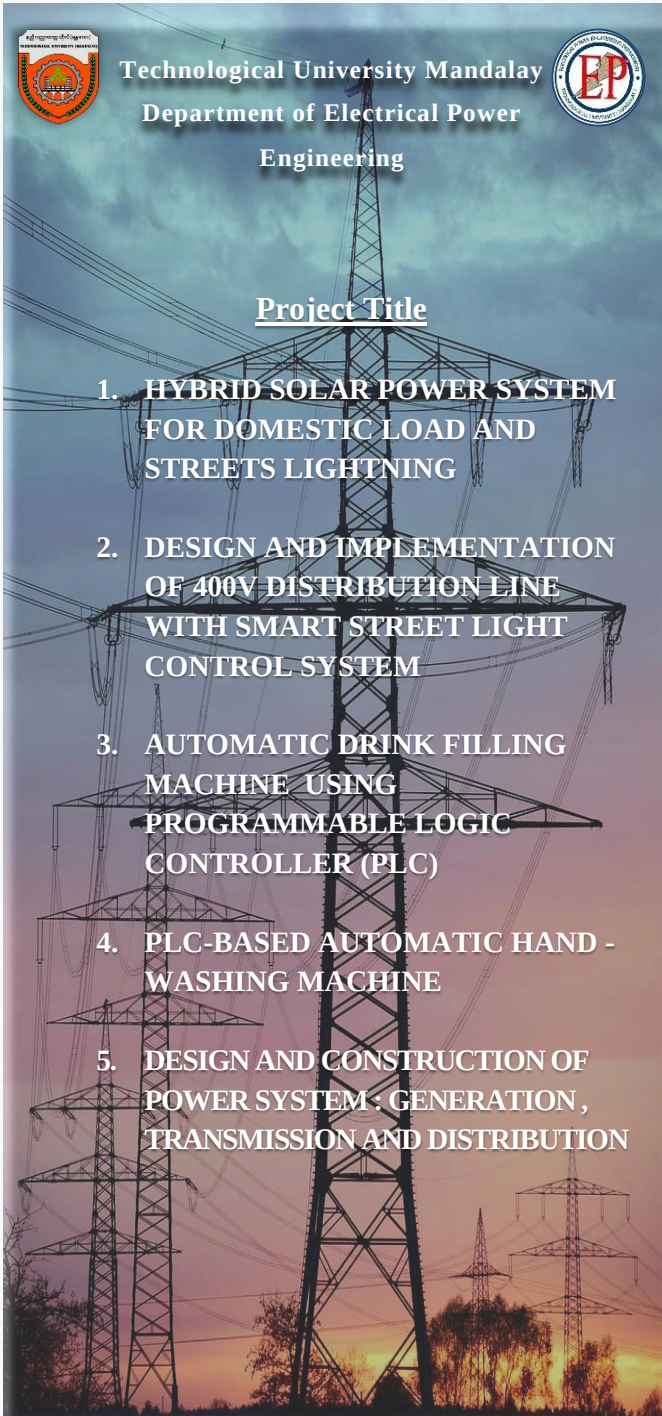


Technological University Mandalay
Department of Electrical Power
Engineering



Project Title

1. HYBRID SOLAR POWER SYSTEM FOR DOMESTIC LOAD AND STREETS LIGHTNING
2. DESIGN AND IMPLEMENTATION OF 400V DISTRIBUTION LINE WITH SMART STREET LIGHT CONTROL SYSTEM
3. AUTOMATIC DRINK FILLING MACHINE USING PROGRAMMABLE LOGIC CONTROLLER (PLC)
4. PLC-BASED AUTOMATIC HAND - WASHING MACHINE
5. DESIGN AND CONSTRUCTION OF POWER SYSTEM : GENERATION , TRANSMISSION AND DISTRIBUTION



DESIGN AND CONSTRUCTION OF POWER SYSTEM: GENERATION, TRANSMISSION AND DISTRIBUTION

This project presents a physical power system model demonstrating electricity generation, transmission, substation transformation, and distribution with LED indicators for visualizing real-time power flow.

Aim

- To design and construct a prototype demonstrating the complete power system from hydro-generation to transmission and residential distribution.

Objectives

- To model hydroelectric power generation using a dam and represent transmission using step-up transformers.
- Demonstrate voltage regulation and step-down conversion for residential power supply.
- Visualize the complete power flow from generation through transmission to distribution.

Methodology

Scale model and Grid → Circuit Wiring and LED → Testing and Power Flow → Result Evaluation



PLC-BASED AUTOMATIC HAND-WASHING MACHINE

This project presents a PLC-based automatic hand-washing machine that uses sensors to control soap, water, hand drying, and gel dispensing without physical contact.

Aim

- To design and implement a PLC-based automatic hand-washing machine for touchless and hygienic hand-washing.

Objectives

- Develop a photoelectric hand-detection system and control the washing process using PLC ladder logic.
- Automate soap, clean-water, hand-dryer, and hand-gel operations with proper stage interlocking.
- Provide battery backup during power failures while improving hygiene and reducing water and liquid consumption.

Methodology

System Design → Component Selection → Mechanical Construction → Electrical Wiring → PLC Programming → System Integration → Testing → Result Evaluation



AUTOMATIC DRINK FILLING MACHINE USING PROGRAMMABLE LOGIC CONTROLLER (PLC)

This project presents the design and implementation of an automated drink filling Machine controlled by a Programmable Logic Controller (PLC) to streamline liquid packaging processes.

Aim

- Design and construct an automated drink filling prototype integrated with PLC control.

Objectives

- Implement precise liquid level control using sensors and solenoid valves.
- Develop PLC ladder logic for automatic cup detection and filling.
- Increase production speed, reduce filling errors, and minimize manual labor.

Methodology

System requirement analysis → Ladder logic programming → Sensor & actuator wiring → Hardware integration → Performance testing & validation.

